

Cyber Security Awareness Booklet



WB-Cyber Security Incident Response Team

Department of Information Technology and Electronics,
Government of West Bengal

A Fun Way to Learn about Cyber Security and Defenses

Executive Summary

This document intends to educate and spread awareness among citizens, government employees, private organizations employees etc. about Cybersecurity. The document is prepared such that this will help everyone to be more cybersecure personally and professionally.

This document is divided into four sections.

PART-A: Infotainment with Cyber Awareness (Information entertainment and Awareness) -

This section is for those who find reading about cybersecurity a bit boring. This section provides cybersecurity awareness in an easy-to-understand way and at the same time making learning a fun.

PART-B: Quick Reference - This section provides a quick reference summary on critical points of cybersecurity and also references to more detailed important CERT-In, Gol guidelines.

PART-C: Incident response – Many times, in general a user is not sure what to do when an cyber incident happens. This section describes what do in such situation along with incident reporting.

PART-D: Glossary - A compilation of key terms and definitions to support understanding of cybersecurity concepts discussed in the document.

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A. Infotainment with Cyber Awareness (Information entertainment and Awareness)

A. Introduction to Cybersecurity: Making It Fun and Easy to Understand

Welcome to the world of Cybersecurity! Think of this as your personal guidebook through the digital jungle, where dangers lurk but can be outsmarted with the right knowledge. So, let's make this journey fun and enlightening!

Why Cybersecurity is Like Being a Superhero

Imagine you're a superhero, and your mission is to protect a hidden treasure – that's cybersecurity in a nutshell! In our digital world, this treasure is your personal and company data. Just as superheroes fight against villains to protect the world, we battle against cyber threats to keep our information safe.

The Superpowers of Cybersecurity

1. **Invisibility Cloak (Confidentiality):** Like having an invisibility cloak, confidentiality means keeping our secret information hidden from prying eyes. Whether it's your secret chocolate cake recipe or company secrets, only the right people should see them.
2. **Truth Spell (Integrity):** This is like casting a spell that ensures nothing gets changed or tampered with – your data stays pure and unaltered, just like the words of a truth spell.
3. **Teleportation (Availability):** Imagine you can teleport anywhere, anytime. Availability in cybersecurity is like that, but for your data. You can access your information whenever you need it, without delay.
4. **Detective Skills (Accountability):** Like a detective, accountability helps us keep track of who did what. If something changes, we can find out who's behind it - just like solving a mystery.
5. **Shield of Protection (Risk Management):** Think of this as your shield against dangers. It's about knowing the risks in the digital world and having the right armor to protect against them.

Fun Facts Corner!

- Did you know the first computer virus was created in 1986 and was called “Brain”? It's like the first super-villain of the computer world!
- The password for the computer controls of nuclear-tipped missiles of the U.S was 00000000 for eight years! Talk about easy to guess!

Conclusion: Becoming a Cybersecurity Superhero

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Now that you've got a glimpse into the world of cybersecurity, remember, every one of us can be a cybersecurity superhero. By understanding these principles, you're already on your way to safeguarding your digital universe!

2. Recognizing Cyber Threats

Welcome to the cyber detective's corner! Here, we'll learn about the sneaky villains of the cyber world and how to spot them. Let's put on our detective hats and start unraveling the mysteries of cyber threats.

Types of Cyber Threats

Cyber threats come in many disguises, just like villains in comic books. Here are some of the most notorious ones:

1. **Phishing Scams:** These are like the disguise artists, tricking you into giving away secret information.
2. **Malware:** This is a sneaky bug that gets into your computer and causes chaos.
3. **Ransomware:** Imagine someone locking up your favorite toy and asking for candy in return. That's ransomware, but with your important data!
4. **DDoS Attacks:** Think of this as a traffic jam blocking the way to your favorite website.
5. **Insider Threats:** Sometimes, the villain might be someone from the inside, like a twist in a detective story.

Story 1: "The Invisible Invader"

Let's meet Ravi, a bright employee at ABC Corp. One day, Ravi received an email with a link saying, "Check out this cool video!" Excited, Ravi clicked the link without a second thought. But little did Ravi know, this click was about to unleash a hidden villain - Malware!



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The malware, like an invisible invader, sneaked into Ravi's computer. It started to gather secret information and even spread to other computers in the network. But how did this happen?

Here's the twist: The email was not from a friend, but from a cyber villain, cleverly disguised. The link was a trap, and the 'cool video' was actually a malware attack in disguise!

The Lessons from Ravi's Adventure

1. **Always Be Suspicious:** If an email looks too good to be true, it probably is. Like a detective, question everything.
2. **Look Before You Leap:** Before clicking on any link, hover over it to see where it leads. Think of it like checking for traps before stepping.
3. **Keep Your Shields Up:** Use antivirus software as your shield. It helps in catching these invisible invaders.

Phishing Scams

Story 2: "The Phishing Expedition"

In the bustling world of ABC Corporation, there lived an eagle-eyed employee named Priyanka. One fine day, Priyanka received an email that seemed to be from the company's CEO, asking for sensitive financial information. But Priyanka, with a detective's instinct, noticed something fishy. The email address had an extra letter, and the message seemed unusually urgent.



Priyanka remembered the training on phishing scams – those tricky emails that try to steal your secrets. Acting swiftly, Priyanka reported the email to the IT department, foiling a crafty phishing attempt. Kudos to Priyanka for being a cyber guardian!

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Lessons from the Expedition

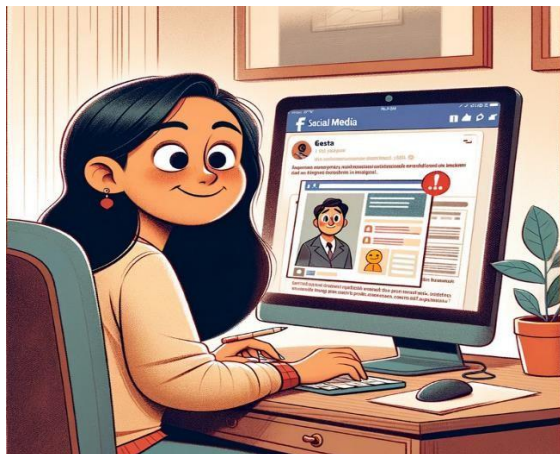
1. **Always Verify Sender Information:** Be cautious about emails from unknown or suspicious sources.
2. **Beware of Urgent Requests:** Phishing emails often create a sense of urgency to provoke hasty actions.
3. **Report Suspicious Emails:** Instead of responding, report any dubious emails to your IT department.

3. Safe Online Practices

Internet and Email Etiquette

Story 3: "The Perilous Post"

Geeta, a well-liked employee at ABC Corp. One day, Geeta decided to share a work-related success on social media, including a screenshot of a confidential report. Uh-oh! This innocent post accidentally exposed sensitive data to the public. Geeta's story became a valuable lesson at ABC Corp about the importance of thinking twice before sharing anything work-related online. Remember, the internet never forgets!



"The Perilous Post" (Geeta's Mistake) Lessons Learnt

- **Think Before You Share:** Always be mindful of what you post on social media, especially if it relates to work.
- **Sensitive Information Stays Off Social Media:** Never share confidential work-related information publicly.
- **Privacy Settings Matter:** Regularly check and update your social media privacy settings.

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Password Security

Story 4: "The Password Predicament"

Manoj had a favorite password: "Manoj123". Easy to remember, right? Manoj used it everywhere – from email to social media to work accounts. But one day, one of these sites got hacked, and Manoj's favorite password was no longer a secret. Soon, Manoj found strange activities in all accounts using "Manoj123". The moral? Use unique, strong passwords for different accounts – it's like having different keys for every door!



"The Password Predicament" (Manoj's Dilemma) Lessons Learnt

- **Use Strong, Unique Passwords:** Avoid using the same password across multiple platforms.
- **Change Passwords Regularly:** Update your passwords periodically to enhance security.
- **Consider a Password Manager:** Use a reliable password manager to keep track of different strong passwords.

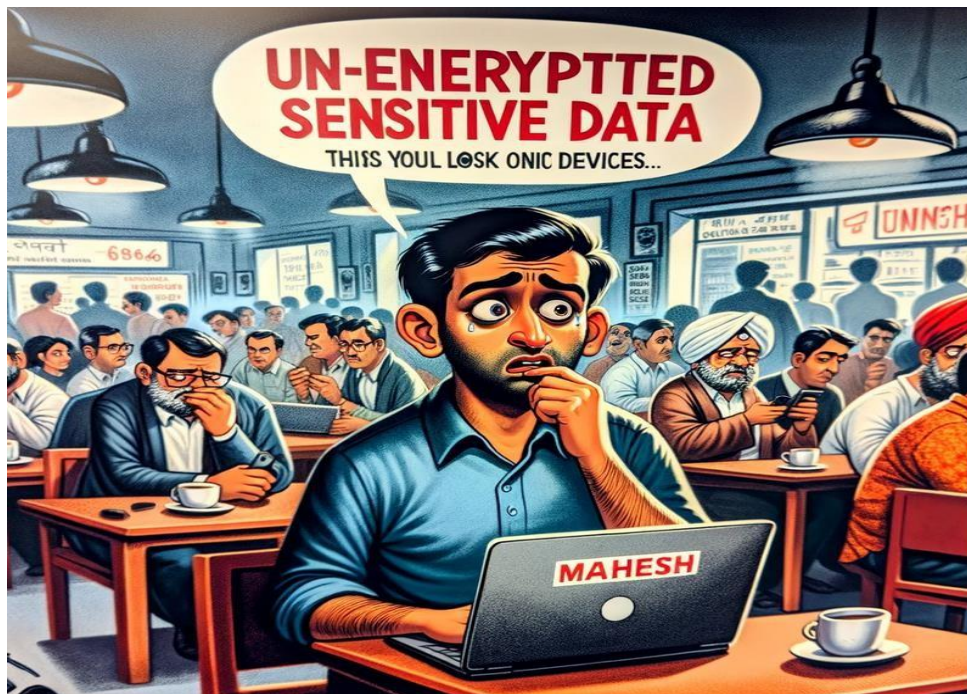
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4. Securing Personal and Company Data

Data Protection Techniques

Story 5: "The Lost Laptop"

Imagine losing a laptop at a café. Now imagine that laptop had unencrypted sensitive company data. This nightmare turned real for Mahesh at ABC Corp. The lost laptop was a wake-up call for everyone. It led to the adoption of strict data encryption policies and secure device management practices. A lost device shouldn't mean lost data!



"The Lost Laptop" (Mahesh's Oversight) Lessons Learned.

- **Secure Your Devices:** Always encrypt sensitive data on laptops and other portable devices.
- **Never Leave Devices Unattended:** Be vigilant about your devices in public places.
- **Backup Important Data:** Regularly backup your data to prevent loss in case of device theft or loss.

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Regular Software Updates

Story 6: "The Outdated Outrage"

Bhavesh, a tech whiz, kept postponing software updates on her computer. "I'll do it tomorrow," she thought. But, an unpatched vulnerability in her outdated software gave hackers a backdoor entry into the network. This incident highlighted the importance of timely updates. Postponing updates can be like leaving your house door unlocked!



"The Outdated Outrage" (Bhavesh's Neglect)

- Regular Updates are Crucial: Keep your software up to date to protect against vulnerabilities.
- Don't Ignore Update Notifications: Treat software update reminders as important, not as a nuisance.
- Understand the Risks of Outdated Software: Recognize that outdated software can be a gateway for cyberattacks.

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5. Responding to Security Incidents

Steps to Take in Case of a Breach

Story 7: "The Ransomware Reckoning"

Ankur clicked on a link in an email, and suddenly, his files were locked, with a message demanding payment to unlock them – a classic ransomware attack. But instead of panicking, Ankur remembered the protocol. He immediately disconnected his computer from the internet and intranet and reported the incident. The IT team swung into action, containing the threat and restoring data from backups. Ankur's quick thinking and the team's preparedness turned a potential disaster into a manageable incident.



"The Ransomware Reckoning" (Ankur's Encounter)

- Be Wary of Unknown Links and Downloads: Avoid clicking on suspicious links or downloading unverified attachments.
- Immediate Response to Threats: Quickly isolate affected systems and inform IT security in case of a suspected attack.
- Do not switch off the computer till evidence for analysis have been taken by authorized team.
- Regular Backups and Recovery Plans: Ensure regular backups of critical data and have a recovery plan in place.

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6. Network and Infrastructure Security

Story: "The Firewall Fortress"

At ABC Corporation, a new firewall was implemented, named 'The Fortress'. One day, an attempt to access the network from an unknown source was detected. Thanks to 'The Fortress', the intrusion was blocked, and the network remained secure.



- **Key Learning:** A robust firewall is essential for guarding against unauthorized network access.
- Regularly update and monitor firewall settings to adapt to evolving cyber threats.
- Implement rules to control inbound and outbound traffic based on business needs.

7. Identity and Access Management

Story: "The Key Keeper"

Ravi, an IT admin at ABC Corporation, implemented a new access management system. When a former employee tried to access the system, the request was denied, as Ravi had promptly revoked their access upon their departure.



- **Key Learning:** Regularly updating access permissions and employing strong identity management protocols are crucial for maintaining system security.

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8. Application Security

Story: "The Secure Code Saga"

Anika, a developer at ABC Corporation, always followed secure coding practices. This habit paid off when a major vulnerability was discovered in applications using similar code, but Anika's application remained unaffected.



- Follow secure coding practices to minimize vulnerabilities.
- Conduct regular security testing during the software development lifecycle.
- Stay updated with the latest security trends and coding standards.

9. Data Security

Story: "The Data Shield"

A new data encryption policy was put in place at ABC Corporation. When a laptop was lost, the encrypted data remained secure, preventing any potential data breach.



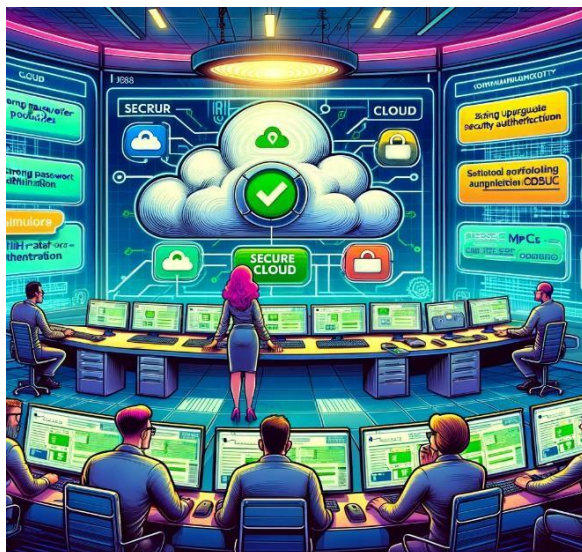
- **Key Learning:** Encrypt sensitive data to prevent unauthorized access.
- Implement data loss prevention strategies to safeguard critical information.
- Regularly back up data to ensure recovery in case of a breach or loss.

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10. Secure Cloud Services

Story: "The Cloud Lockdown"

ABC Corporation's IT team conducted a thorough review of their cloud services, tightening access controls and implementing MFA. This proactive step thwarted a potential breach through a cloud service.

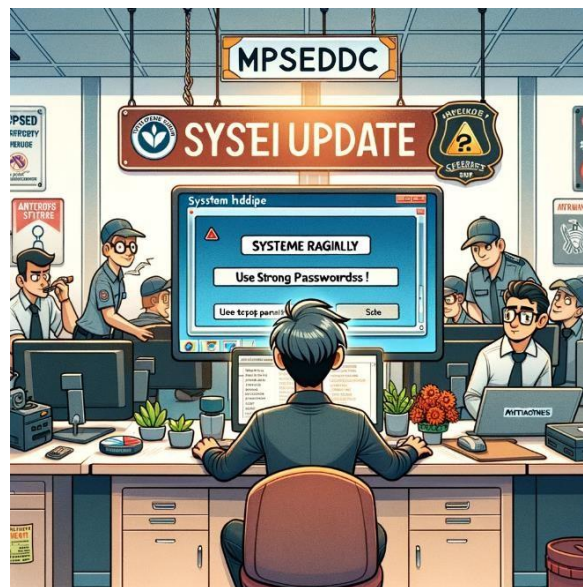


- **Key Learning:** Continuously monitor and review cloud services for potential vulnerabilities.
- Implement strong password policies and multi-factor authentication for cloud access.
- Ensure that cloud configurations do not inadvertently expose sensitive data.

11. Hardening Procedures

Story: "The Secure Setup"

After implementing system hardening guidelines, ABC Corporation noticed a decrease in security incidents. Restrictions on unauthorized software and regular system updates played a key role.



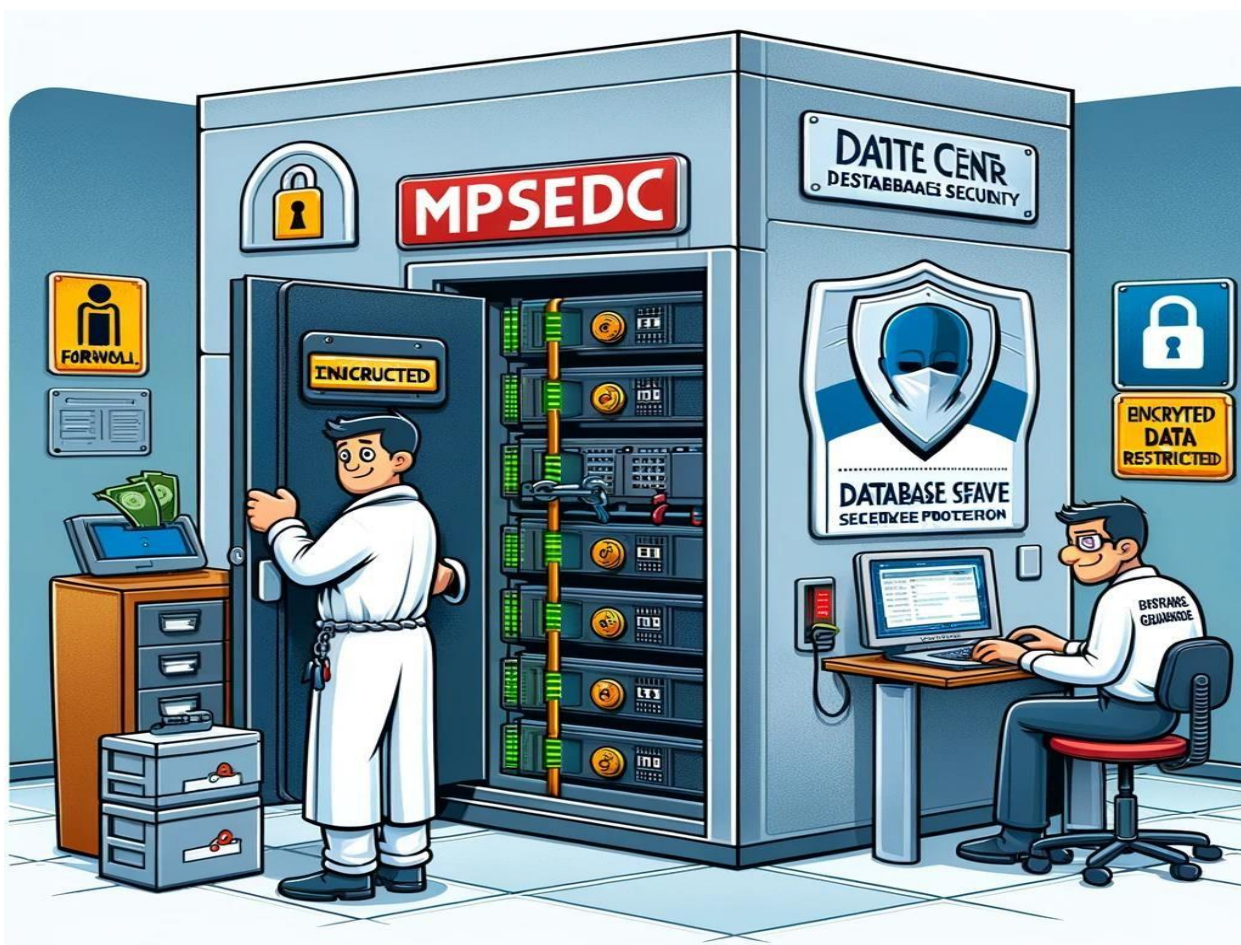
- **Key Learning:** Use genuine and supported software to avoid vulnerabilities found in pirated versions.
- Regularly update all systems and software to patch security vulnerabilities.
- Limit administrative privileges and use standard user accounts for daily operations.

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12. Database Server Security

Story: "The Database Guardian"

A new policy at ABC Corporation ensured that their database servers were isolated and protected by advanced firewalls. This move successfully blocked an attempted SQL injection attack.



- **Key Learning:** Isolate database servers and restrict access with firewalls and access controls.
- Regularly update database management systems for security patches.
- Use complex passwords and monitor for suspicious login activities.

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Conclusion of the Cybersecurity Awareness and Training Manual

As we conclude our journey through the cybersecurity landscape at ABC Corporation, we hope this manual has not only enlightened but also empowered you. Cybersecurity is not just a responsibility of the IT department; it's a shared commitment across all levels of the organization.

Embracing a Culture of Cybersecurity

- **Awareness is Key:** Regular awareness and training are crucial. Stay updated with the latest threats and safe practices.
- **Every Action Counts:** Each one of us plays a vital role in safeguarding our organization's digital assets. Even small actions can have a big impact on our collective security.

Implementing Best Practices

- Implement the lessons learned from the stories and guidelines in this manual. From 'The Firewall Fortress' to 'The Database Guardian', each story has imparted valuable insights into protecting our digital environment.
- Regularly update your passwords, be vigilant about email security, and always think twice before clicking on links or sharing sensitive information.

Moving Forward

- Cybersecurity is an evolving field. What is secure today may not be tomorrow, so continuous learning and adaptation are essential.
- Encourage open communication about cybersecurity issues. If you see something suspicious, say something.

Together, We Secure

- Remember, cybersecurity is a team sport. We are all in this together, and it is our collective efforts that will keep ABC Corporation secure.

This manual is not just a one-time read but a guide to be revisited regularly. Let's commit to making cybersecurity a part of our daily routine and culture at ABC Corporation. Together, we can create a safer digital space for ourselves and our organization. Stay vigilant, stay informed, and stay secure.

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B. Quick Reference

Security Policy: Formulate security policies and procedures for building cyber resiliency. Prepare, test and implement Business Continuity Plan (BCP) and Disaster Recovery (DR) plan.

Cyber Security Awareness: Prepare an organization-wide Cyber Security Awareness Program and regularly educate end users

Network segmentation: Network should be properly segmented, with separate VLANs for different functional requirements.

Firewall and monitoring tools: Use firewalls to create a buffer zone between the Internet (and other untrusted networks) and the networks used by the organization. Have monitoring tools such as IDS/IPS. Allow only valid connections.

Email security: Deploy web and email filters on the network. Configure these devices to scan for known bad domains, sources, and addresses; block these before receiving and downloading messages. Scan all emails, attachments, and downloads both on the mail gateway and hosts with a reputable antivirus solution.

Logging and monitoring: All devices placed within the network should have logging enabled. Logs must be monitored for anomalies.

Access through authorized machines: Implement Media Access Control (MAC) address binding for all systems/IT devices. Disable DHCP and set IP configurations manually.

Default settings: Change all default credentials & configurations at the time of first installation.

Disable unused ports/services: The organization must identify ports, protocols and services required to carry out daily operations and block all others, including all non-IP based and unencrypted protocols, by establishing policies in routers and wireless access points.

Least access: Access privileges to users must be based on operational role and requirements. All user accounts must be reviewed periodically and must have least privilege, just enough required to do the job requirements.

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Complex password: Use a complex password and it shall not be same on different applications. Password should not have a pattern, Ex Abcd1234! Is a very easy password.

Applications list: The organization must maintain an updated document containing the list of authorized applications, their usage, custodian(s) assigned to each application.

Encryption: Ensure that all Websites and Applications are “https” enabled with a valid SSL/TLS Certificate.

Data classification: Identify and classify sensitive/personal data and apply measures for encrypting such data.

Back-up: Back-up procedures should be documented, scheduled and monitored. Proper back-up is the only effective way to deal with Ransomware attack.

NDA with third party: Organization should ensure that third party access to information should be restricted and should only be shared after signing Non-Disclosure-Agreement.

Genuine software: Ensure that all office systems are installed with genuine copy of operating system and other application software solutions. It is recommended to install only whitelisted software on office systems based on the roles of the users.

No wireless on sensitive systems: All the terminals or computers dealing with sensitive/classified information should not have any wireless equipment including Internet and Bluetooth.

Software update: The organization must ensure that patch management is carried out at regular intervals or as soon as critical patches for ICT systems or software are available.

Audits: Security audit should be performed prior to and after implementation or installation or major enhancements in the organization. 3 rd Party Security audits must be conducted periodically at least once a year.

Incident response plan: Each organization must have an incident response plan to take timely and effective actions incase of Cyberattack.

Guidelines: All government entities must follow **guidelines issued by CERT-In, Government of India** @ <https://www.cert-in.org.in/PDF/guidelinesgovtentities.pdf>

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C. Incident response for a common user/organization

**** Pre planning: Each Organization must have a incident management plan, and incident management team with proper roles and responsibilities.**

Important steps to be taken when an organization is affected by cyber attack.

1. Inform your incident response team.
2. Identify compromised systems.
3. Disconnect the compromised system from the internet and Intranet, **but don't turn it off**. Turning of the system could result in loss of crucial evidence which would be needed for the analysis and investigation of the incident.
4. Report the data breach/ data leak/ Cyber-attack to law enforcement agencies such as Cyber cell police, CERT-In Incident Response Help Desk immediately. CERT-In must be informed about incident within 6 Hours at email id incident@certin.org.in.
5. Let authorized and technically competent team take digital evidences for forensic investigation and generate hashes for primary and back-up data state.
6. Log collection for review and analysis.
7. Let incident response team invoke incident management plan.
8. Check unaffected systems for any unusual activity, connection. Scan with latest updated antivirus, scan for IOC (Indicators of Compromise).
9. As applicable, notify users/customers who could be affected immediately with details of information breached; actions being undertaken to address the

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problem and how they can reach back for any queries. Users shall also be informed to update the password.

10. Ensure all credentials in an organization, including service accounts, are reset.
11. Update signatures of IDS (Intrusion Detection System), Antivirus, Firewall rules etc. as applicable.
12. After confirmation from incident response team / authorized team, sanitize the system/ rebuild the system and upload data from identified clean back-up, as per situation.
13. Analyze logs, malware, IOC/IOA (Indicators of Attack) etc. to know root cause of attack and path of attack.

Common Post Incident Activities

1. Document incident, action taken, learnings etc.
2. Prepare report for stakeholders.
3. Identify changes / finetune processes / controls as needed.
4. Keep close watch on systems after recovery.
5. Awareness session and learnings within organization may be done.
6. Re-audit may also be done, incase of major incidents.

Incident Response Phases

1. **Preparation:** Establishing and training the response team, setting up tools and processes to prevent security incidents.
2. **Identification:** Detecting and determining the nature and scope of the incident.
3. **Containment:** Isolating affected systems to prevent further damage or spread of the incident.
4. **Eradication:** Removing the threat from the environment, including deleting malicious files and disabling breached user accounts.
5. **Recovery:** Restoring and returning affected systems and devices back to their fully operational states.

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6. **Lessons Learned:** Reviewing and analyzing the incident response process and outcomes to improve future responses.

Easy to understand flow of Incident Response

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Incident Reporting:

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- <https://cybercrime.gov.in/> : Filing a Complaint on National Cyber Crime Reporting Portal. Call 1930 for financial fraud (moving flag of website)
- <https://cybercrimewing.wb.gov.in/> : WB State Cyber Crime Wing
- <https://ceir.gov.in/Home/index.jsp> : Central Equipment Identity Register to report lost or stolen mobile phone
- incident@cert-in.org.in : Communication about cyber security incident report

D. Glossary

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1. **Cyber Security:** The practice of protecting systems, networks, and programs from digital attacks.
2. **Phishing Scams:** Fraudulent attempts to obtain sensitive information by disguising oneself as a trustworthy entity in an electronic communication.
3. **Malware:** Malicious software designed to harm or exploit any programmable device, service, or network.
4. **Ransomware:** A type of malicious software designed to block access to a computer system until a sum of money is paid.
5. **DDoS Attacks:** Distributed Denial of Service attacks that overwhelm a system's resources, making it unavailable to users.
6. **Insider Threats:** Security threats that come from people within the organization, such as employees, former employees, contractors, or business associates.
7. **Confidentiality, Integrity, Availability (CIA):** Fundamental principles of information security.
8. **Risk Management:** The process of identifying, assessing, and controlling threats to an organization's capital and earnings.
9. **Incident Response Plan:** A set of instructions to help IT staff detect, respond to, and recover from network security incidents.
10. **Encryption:** The process of converting information or data into a code, especially to prevent unauthorized access.
11. **Firewall:** A network security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.
12. **Intrusion Detection System (IDS)/Intrusion Prevention System (IPS):** A device or software application that detects/monitors/captures a network or systems for malicious activity or policy violations.
13. **Multi-Factor Authentication (MFA):** A security system that requires more than one method of authentication from independent categories of credentials to verify the user's identity for a login or other transaction.
14. **Indicators of Compromise (IOC):** Signs that suggest a security breach has occurred, like unusual files or suspicious network activity.
15. **Indicators of Attack (IOA):** Early warning signs of an ongoing attack, focusing on the behavior and techniques used by attackers.